

Business Requirement Document (BRD)

Power BI Project – TAT (Turn Around Time) Analytics POC

1. Document Control

- **Project Name:** TAT (Turn Around Time) Analytics – POC
 - **Client / Business Unit:** Diagnostics / Lab Operations
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2. Business Background

The organization wants to analyze **Turn Around Time (TAT)** performance across laboratories, departments, tests, and geographies. The goal of this POC is to identify operational bottlenecks, measure SLA adherence, and improve efficiency by tracking whether tests are completed **within defined TAT thresholds**.

The analysis will help stakeholders understand: - Overall TAT performance - Variations by lab, department, test, and geography - Sample journey delays across different processing stages

3. Business Objectives

- Monitor overall test volumes and TAT compliance
 - Identify labs/departments with poor TAT performance
 - Understand rejection, re-run, hold, and outsourcing impact on TAT
 - Analyze hourly load and sample movement delays
 - Enable data-driven operational improvements
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4. Scope of the Project

4.1 In Scope

- Development of Power BI dashboards for TAT analysis
- KPI calculation based on provided TAT dataset
- Comparative analysis across dimensions
- Hourly sample journey analysis
- POC-level data modeling and visualization

4.2 Out of Scope

- Predictive analytics / ML
 - Real-time streaming dashboards
 - Data corrections at source systems
 - User-level security (RLS) for POC
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5. Key Stakeholders

- Business Operations Team
 - Lab Operations Managers
 - Department Heads
 - Senior Management
 - Power BI Development Team
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6. Data Sources

Source	Description
TAT Transaction Data	Test registration, processing, and completion timestamps
Master Data	Lab, Department, Test, City, State

Data will be provided in Excel / CSV / Database format for POC.

7. Key Business Definitions

- **TAT (Turn Around Time):** Time taken from test registration to final approval
 - **In TAT:** Test completed within defined SLA
 - **Out TAT:** Test completed beyond SLA
 - **% In TAT:** ($\# \text{ Tests In TAT} / \# \text{ Total Tests Performed}$) * 100
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8. KPIs & Metrics (As Per Requirement)

8.1 Overall KPIs

· **Total Tests Performed**

· **Tests Outsourced**

· **Tests Rejected**

· **Tests on Hold**

· **Tests Re-Run**

· **Tests In TAT**

· **Tests Out of TAT**

- % In TAT
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8.2 Dimension-wise KPIs

Lab-wise

- % In TAT by Lab

Department-wise

- % In TAT by Department

State & City-wise

- % In TAT by State
- % In TAT by City

Test-wise

- % In TAT by Test Type
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8.3 Comparative Analysis

- Lab vs Lab – % In TAT comparison
 - Department vs Department – % In TAT comparison
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8.4 Sample Journey & Load Pattern Analysis

Hourly slab-wise analysis of sample movement: - Registration → SRA Receive - SRA Receive → Department Receive - Department Receive → Result Entry - Result Entry → Approval

Purpose: - Identify peak load hours - Identify delay-prone stages

9. Dashboard Requirements

9.1 Summary Dashboard

- KPI Cards (Total Tests, % In TAT, Rejected, Outsourced, etc.)
- Trend of % In TAT over time
- In TAT vs Out TAT distribution

9.2 Performance Analysis Dashboard

- Lab-wise % In TAT
- Department-wise % In TAT
- Test-wise % In TAT

9.3 Geography Dashboard

- State-wise % In TAT (Map / Bar)
- City-wise % In TAT

9.4 Sample Journey Dashboard

- Hourly volume trends
 - Stage-wise delay visualization
 - Funnel / Sankey for sample journey
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10. Filters & Slicers

- Date Range
- Lab
- Department
- Test Type
- State

- City
- In TAT / Out TAT

All visuals should interact dynamically with slicers.

11. Data Model Design (High Level)

11.1 Fact Table

- Test Transactions
- Test ID
- Registration DateTime
- SLA TAT
- In TAT Flag
- Stage Timestamps

11.2 Dimension Tables

- Dim Lab
- Dim Department
- Dim Test
- Dim Date
- Dim Geography (State, City)

Star schema with **single-direction filtering**.

12. Data Refresh Strategy

- Manual refresh for POC
 - Daily refresh (recommended for production)
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13. Assumptions

- Data provided is accurate and complete
 - SLA rules are clearly defined
 - One test belongs to one lab and department
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14. Risks & Mitigations

Risk	Mitigation
Missing timestamps	Data validation rules

Risk	Mitigation
Incorrect SLA mapping	Business sign-off
Large data volume	Aggregations / optimization

15. Success Criteria

- Accurate calculation of all KPIs
- Clear identification of bottlenecks
- Positive stakeholder feedback on usability
- POC delivered within timeline

16. Deliverables

- Power BI .pbix file
- KPI definition document
- Data model diagram
- POC walkthrough session

17. Approval & Sign-off

Name	Role	Signature	Date
	Business Owner		
	IT Owner		

End of Document